

PRODUCT REQUIREMENTS DOCUMENT

Sales Performance Dashboard

A web application for exploring, comparing, and asking questions about sales data.

Product direction: let anyone upload sales-related data and choose how to understand it - interact with a dashboard, build an analysis, see an automatic overview, or ask the data in chat.

Document field	Current definition
Document	Product Requirements Document (PRD)
Project name	Sales Performance Dashboard
Product type	Interactive web application
Intended audience	Anyone who wants to analyze sales-related data
Status	Finalized initial-build requirements; Streamlit stack, repository layout, and hosting path are selected; future cloud integrations remain out of scope
Build constraint	A five-hour first build was requested; the PRD separates a small demonstrable MVP from the complete product vision.

How to use this PRD

This document describes the agreed product direction and its requirements. It does not mean the features have already been built or tested. The full vision contains more than a five-hour build can reasonably deliver, so the MVP and later-phase features are identified separately.

A note about “any data”

The product is for any user, but the uploaded dataset must contain sales-related information. It should adapt to the columns a file has and explain when the file cannot support a requested analysis.

1. Product overview

1.1 Problem statement

People receive sales data in different files, but turning it into useful results can require spreadsheet skills, manual cleanup, and repeated chart building. Users need one approachable website that can inspect sales data, show what it contains, and let them explore it in the way that suits them.

1.2 Product goal

Help anyone upload sales-related data and analyze it through an interactive dashboard, guided analysis, an automatic overview, or a chat interface. Results must reflect the uploaded data, respond to user selections, and be easy to check and present.

1.3 Product principles

- Adapt to the fields in the uploaded file; do not expect every company to use identical column names.
- Make the dashboard interactive: selecting a time period or category updates the relevant KPIs, charts, tables, and insights.
- Show exact numeric answers from calculations. Do not invent values when the data is missing or unclear.
- Keep the chat for questions and short text replies; put requested charts, tables, and maps in the main results area.
- Explain data-quality changes and show the filters or periods behind a result.
- Keep the first screen concise and welcoming rather than presenting a long feature list.

1.4 Target users

The product is not limited to one profession or experience level. It is for students, analysts, business owners, and other people who want to understand sales data, including users with limited data-analysis experience.

1.5 Success outcomes

- A first-time user can understand the four analysis paths and upload a supported sales file.
- The dashboard accurately calculates and refreshes results when the user changes a date range, filter, or selected chart item.
- A user can ask a supported question and get a traceable answer and a relevant result on the main screen.
- The app clearly identifies unavailable analyses rather than displaying misleading or fabricated results.

2. User experience and main paths

2.1 Home page

The first page introduces the website in a few lines. It presents four concise option cards, visible together on a computer and swipeable on a phone. A clear “Get Started” action begins the selected path.

Option card	What the user can do
Explore the dashboard	Upload a file and inspect interactive charts, KPI cards, and filters.
Build an analysis	Use a search box and dropdown choices to pick a question, measure, grouping, comparison, and result format.
Automatic overview	Upload a file and receive the available dashboard analysis and data-grounded insights.
Ask in chat	Ask a sales question in plain language; review the answer and request a visualization.

2.2 Standard flow

- Open the website and read the short introduction and four option cards.
- Choose an analysis path and upload a sales file, or select a sample dataset to preview the experience.
- The app checks the file, identifies likely columns, and shows a preview and data-quality summary.
- The user confirms or corrects the column matches when necessary.
- The selected analysis opens in the main results area.
- The user explores, asks questions, changes filters, views underlying data, and exports or shares results when those capabilities are available.

2.3 Interactive dashboard behavior

The dashboard shows all available data by default. A user may filter by a field such as date, country, state, region, product, or category. Clicking a chart item can filter related dashboard visuals. Selecting January should refresh the relevant totals, charts, tables, and insights to January data; selecting October should refresh them again. A clear/reset action returns to the full dataset.

Date controls only appear when a usable date field exists. Other filters and analyses appear only when the corresponding uploaded fields are available.

2.4 Result area and chat

The chat panel is the place to ask questions and receive short text responses. Charts, tables, and maps appear in the main dashboard/results area, not embedded as large chat bubbles. If a user asks a question that can be answered in text, the app may ask whether they want a visualization and offer suitable chart choices. If the user explicitly asks for a chart, render it directly without an unnecessary confirmation.

3. File upload and data preparation

3.1 Supported input formats

- CSV files.
- Excel workbooks (.xlsx); when there are multiple sheets, let the user choose and preview a sheet.
- TSV (tab-separated) files.

Cloud-drive selection is a later decision. The first upload flow should work with a file selected from the user's device.

3.2 Flexible data shape

There is no single fixed list of columns that every sales file must contain. The app should identify available sales-related fields and enable the analyses they support. At minimum, it must recognize enough sales-related information to perform a meaningful calculation; otherwise, explain what is missing and allow the user to correct column mappings.

3.3 Column recognition

- Suggest likely matches for date, sales/revenue, profit, order ID, quantity, customer, product, category, country, state, city, region, returns, ship date, and ship mode.
- Use both column names and sample values when suggesting a match.
- Show confidence and request user confirmation when a match is uncertain.
- Allow the user to manually map, ignore, or change a column.

3.4 Data-quality review and safe cleaning

- Preview a sample of rows and report input row count.
- Report missing values, placeholders such as #N/A, invalid dates or numbers, possible duplicates, blank columns, and fields that could not be mapped.
- Do not delete a row just because an optional field is blank.
- Treat exact duplicate rows separately from repeated order IDs or repeated product lines.
- Keep negative profit values as valid data unless the user identifies a data problem.
- Explain which rows or values were excluded from a particular calculation and why.
- Offer a cleaned-data download in a later-phase export flow.

3.5 Unsupported or unclear files

For empty, unreadable, unrelated, or insufficiently structured files, show a short plain-language message with a next action. Never present an unrelated file as a valid sales analysis.

4. Dashboard and analysis requirements

4.1 KPI cards

KPI	Definition / required fields
Total Sales / Revenue	Sum of the mapped sales amount.
Total Profit	Sum of mapped profit; may include negative values.
Total Orders	Distinct order IDs, if available; otherwise do not label line count as orders.
Total Quantity	Sum of quantity, if available.
Total Customers	Distinct customer IDs, if available.
Average Order Value	Total sales divided by distinct orders; only when order IDs are available.
Profit Margin	Total profit divided by total sales; only when both are available and sales are nonzero.
Growth %	Percentage change between clearly named comparable periods; only when dates and periods support it.

4.2 Required analysis areas

- Time: monthly, quarterly, yearly trends and growth comparisons when dates are available.
- Products: top and bottom performers by sales, profit, or quantity.
- Categories: category and sub-category comparisons.
- Geography: country, region, state, and city comparisons as supported by available location fields.
- Customers: unique customers, top customers, and segment analysis where fields exist.
- Returns: use the return indicator or return amount available in the file; disclose gaps or incomplete coverage.
- Shipping: ship modes and elapsed time from order date to ship date when both dates are available.
- Sales versus profit: show positive and loss-making results and make the selected measure clear.

4.3 Chart chooser

Provide a broad chart chooser for the dashboard and for chat-requested results. The intended catalog includes KPI cards, bar and column charts, grouped and stacked bars, line and area charts, pie and donut charts, combo charts, scatter plots, treemaps, heatmaps, waterfall charts, histograms, maps, and tables. Show only chart types that fit the fields and question; explain when a chart is unavailable or unsuitable.

4.4 Comparisons, drill-down, and controls

- Compare time periods, products, states, and regions side by side.
- Allow drill-down such as region to state to city when the data hierarchy exists.
- Allow users to choose top/bottom item count and ranking measure.
- Allow users to choose date range and time grouping.
- Allow chart style changes and currency-symbol selection when currency is not specified by the file.
- Allow a user-entered sales target and show actual versus target.
- Support readable colors including user-selected positive/negative colors when requested; maintain sufficient contrast.
- Let a chart click filter other dashboard visuals and offer a one-click reset.
- Provide a table view of the records or aggregates behind a chart.

5. Chat, automatic insights, and outputs

5.1 Chat questions

- Accept plain-language questions about metrics, products, categories, locations, customers, returns, shipping, and time periods.
- Support comparisons and follow-up questions within the current dataset context.
- Provide example question prompts for first-time users.
- Interpret requested chart type, date range, grouping, and colors where supported.
- Answer from calculated results, not unsupported free-form guesses.

5.2 Chat-to-dashboard behavior

- For a text question, show a concise text answer in chat.
- For a requested chart, map, or table, place the result in the main dashboard area.
- If the user asks “Which state had the highest sales?”, answer in chat and offer chart choices; after selection, show the chart on the main screen.
- If the user already asks “Show this as a bar chart,” render a bar chart on the main screen directly.
- Include a brief result context: metric, active filters, date range, and fields used.
- Show which uploaded fields and calculations support the answer, with a way to inspect the underlying data.
- When a question is ambiguous, ask one short clarifying question. When the file lacks required information, explain what is missing.

5.3 Automatic insights and alerts

- Show concise findings alongside charts and KPI cards, such as a top-performing state, product, or a period change.
- Flag unusually large or small transactions for review and notable changes over time.
- Support every numeric insight with the same deterministic calculation shown in the dashboard.
- Do not claim an event caused a sales change unless the data proves that relationship.

5.4 Outputs and sharing

- Download a PDF report with user-selected charts, KPIs, and date range.
- Download cleaned data and question-result tables as CSV or Excel.
- Compare two compatible uploaded files, such as this year versus last year; combine compatible monthly files.
- Let users choose whether customer names are included in reports.
- Provide a way to save/share a dashboard. Cloud hosting, persistent saved dashboards, and share-link permissions need a separate privacy and hosting decision.

6. Reference dataset and validation notes

The supplied SuperStore_Sales_Dataset.csv is the initial reference file. It contains 5,901 rows and data from 2019-01-01 through 2020-12-31. It includes order and ship dates, order IDs, customer identifiers and names, country, state, region, city, product, category, sub-category, sales, profit, quantity, payment mode, ship mode, and a returns indicator.

Reference observation	Product implication
3,003 distinct Order IDs across 5,901 rows	Rows are order/product lines; count distinct IDs for Total Orders.
49 states, 4 regions, 3 categories, 17 sub-categories	State, region, category, and sub-category analysis are possible for this sample.
Raw Sales total: 1,565,804.32	Reference calculation only; the file does not declare a currency.
Raw Profit total: 175,262.1059	Keep negative profits; loss-making rows may be meaningful.
Raw Quantity total: 22,317	Quantity KPI can be validated against this total.
Returns is #N/A on 5,614 rows; remaining values are 1	Returns analysis is incomplete; state this limitation visibly.
Two columns are blank; first row-ID header is malformed	Flag/ignore unused or malformed columns rather than treating them as required fields.
Only United States country data; dates cover 2019-2020; no Toys category	The sample cannot answer India-versus-US, 2022-2026, or toy-sales questions. Explain missing coverage; never fabricate.

6.1 Calculation and currency rules

- Show monetary values without a currency symbol unless the user selects one or the source declares it.
- Growth = (current period value - comparison period value) / comparison period value. Name both periods; handle a zero or missing comparison denominator safely.
- Average order value = total sales / distinct orders.
- Profit margin = total profit / total sales.
- Shipping duration = ship date minus order date; show the date-unit definition.

6.2 Reference-file acceptance checks

- The app reports 5,901 input rows and identifies the date and core sales fields correctly or requests correction.
- Unfiltered totals reconcile to the included records, with any exclusions explained.
- Total orders count distinct Order IDs, not row count.
- The app identifies United States coverage and 2019-2020 dates and does not claim India or 2022-2026 results.
- The returns limitation and blank/malformed columns appear in the quality summary.

7. Quality, privacy, and five-hour MVP

7.1 Non-functional requirements

- Beginner-friendly labels and short explanations; responsive layout for phone and desktop.
- Reliable, consistent calculations and clear KPI definitions.
- Handle malformed files without crashing; provide actionable messages.
- Use accessible contrast, readable labels, keyboard-operable controls, and distinguishable chart states.
- Avoid logging uploaded row contents or customer names. Use temporary processing for the first local-file MVP, provide a clear/reset action, and decide retention timing before adding cloud storage or shared links.

7.2 Five-hour demonstrable MVP

The full vision is larger than a five-hour first build. The recommended time-boxed MVP prioritizes a working end-to-end demonstration. The selected five-hour stack is described in Section 8.3. The repository will still separate frontend and backend code into clear folders inside one Python app.

Build priority	Five-hour MVP target
P0 - must demonstrate	Landing page with four mode cards; upload CSV, XLSX, and TSV; preview and column mapping; data-quality summary; responsive interactive dashboard; core KPIs; date and available-field filters; common chart choices; automatic overview; a chat-style query panel with a small set of supported question types; result shown on the main canvas; grounded answer and missing-field messages.
P1 - if time remains	PDF report export; cleaned CSV export; several automatic insights; chart/table switch; top/bottom rankings; basic drill-down; sample data and example questions.
Later phase	Cloud-drive connectors, persistent share links/permissions, multi-user access, robust multi-file combining, advanced visual catalog, advanced anomaly detection, forecasting, and open-ended AI analysis.

7.3 Five-hour build sequence

Time box	Checkpoint
0:00-0:30	Create the app shell and landing page; choose and install the minimum framework/libraries.
0:30-1:15	Implement file upload and parsing for the agreed MVP formats; show useful upload errors.
1:15-2:00	Preview rows, detect/map columns, and report data quality.
2:00-3:15	Build the dashboard: KPI cards, charts, date/filter interactions, and reset.
3:15-4:00	Add automatic overview and a limited, deterministic set of chat questions that update the main results area.
4:00-4:30	Polish responsive layout; check the supplied CSV and a small sample file; fix high-impact issues.
4:30-5:00	Push the code to GitHub and deploy from that repository to Streamlit Community Cloud if the MVP is ready.

If the five-hour limit is strict, a focused working slice is safer than claiming every advanced feature is complete. The remaining accepted features stay in the product backlog for follow-up.

7.4 Repository folder layout

Use one GitHub repository with clear frontend and backend folders. With Streamlit, these folders organize the code responsibilities; they are not separately running services.

```
sales-performance-dashboard/
|-- frontend/           # Streamlit pages and visual components
| |-- app.py           # Entry point selected for deployment
| |-- pages/           # Dashboard, guided analysis, chat
| |-- components/       # Upload, KPI, filter, chart components
| |-- styles/           # Theme and styling
| |-- assets/           # Images and icons
|-- backend/            # Data work imported by the app
| |-- file_handling/    # Read CSV, Excel, TSV
| |-- cleaning/         # Quality checks and safe cleaning
| |-- column_mapping/   # Match uploaded columns
| |-- analytics/        # KPIs, trends, comparisons, insights
| |-- chat/             # Supported data questions
| |-- exports/          # PDF and data downloads
|-- data/sample/        # Safe demo data only
|-- docs/               # PRD and project notes
|-- tests/              # Calculation and upload checks
|-- .streamlit/config.toml # Streamlit settings
|-- requirements.txt     # Python packages
|-- README.md           # Setup, features, and usage
`-- .gitignore           # Exclude secrets and private uploads
```

Do not commit private customer or sales uploads to GitHub. Keep only a safe sample dataset in data/sample/.

8. Acceptance criteria and open decisions

8.1 Product acceptance criteria

- A first-time user can identify the four available analysis paths from the home page and start with one.
- CSV, Excel, and TSV uploads are accepted when valid; bad or non-sales files receive understandable feedback.
- Column suggestions can be reviewed/corrected; a quality report precedes analysis.
- The dashboard shows only analyses supported by the file and updates when date or other filters change.
- Clicking a chart item can filter relevant dashboard results; reset restores the full dataset.
- KPI formulas, comparison periods, and currency assumptions are clear.
- Chat numeric answers match dashboard calculations; requested visual results appear on the main screen, not as a chart embedded in chat.
- The app explains missing fields and out-of-range periods instead of inventing results.
- PDF/data exports, when included in the build, reflect current chosen filters and customer-name settings.

8.2 Decisions recorded

- Project name: Sales Performance Dashboard.
- Audience: anyone analyzing sales-related data.
- Website landing page: four concise cards and a Get Started action; cards are visible together on desktop and swipeable on phones.
- Analysis paths: interactive dashboard, guided analysis, automatic overview, and chat.
- Upload formats selected: CSV, Excel, TSV.
- Dashboard starts with the full dataset; filters are optional.
- User-selected chart options, chart/table views, downloads, insights, and chat-to-dashboard results are part of the product vision.

8.3 Selected five-hour stack and deployment

- Develop the files in Antigravity in a local project folder.
- Use Python and Streamlit for one app. Keep frontend screens in frontend/ and backend data services in backend/; run the app from frontend/app.py.
- Use pandas for CSV, Excel, and TSV reading, cleaning, and calculations; use Plotly for interactive charts.
- Use no database in the first version because it does not save user accounts or dashboards. Process uploaded files for the current app session.
- Store the source code in GitHub, then connect the repository to Streamlit Community Cloud and select frontend/app.py as the app entry point. The host provides the online app link.
- Use free tiers for the first demonstration and avoid a paid AI API; free plans have usage limits that may change.

8.4 Items still needing a technical/product choice

- Cloud-drive import (such as Google Drive) and persistent shared dashboards are later features; decide storage, privacy, and access rules before adding them.
- Which exact chat question types to support first. Start with a fixed set and expand later.
- Maximum file size and row count; set limits appropriate to the deployment host.
- Exact PDF report layout and whether it fits the five-hour build.